

# Hindsight-Guided Rationale Distillation for Rare Disease Diagnosis

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## Abstract

We study hindsight-guided distillation for rare disease diagnosis on ZebraMap: a 1.5B student is fine-tuned on chain-of-thought traces from a 8B teacher that observes the ground-truth diagnosis during generation. Absolute accuracy remains low for all models—the task is hard at this scale—but within this ceiling a filtered variant (StudentF) achieves a small, statistically significant accuracy advantage over the teacher ( $p < 0.001$ ), concentrated in better-represented diseases. The unfiltered student does not significantly outperform the teacher ( $p = 0.129$ ), establishing that contamination filtering—not hindsight distillation alone—drives the gain.

The gap traces to an artifact we term GT hallucination. Label-visible generation causes the teacher to embed “ground truth is X” phrases in its reasoning chain; SFT copies the pattern. At inference, the unfiltered student reproduces the phrase in 33.9% of cases, with severe accuracy degradation when the hallucinated label is wrong. A regex filter removing these slots reduces contamination to near-zero, producing the observed gain—though the effect remains small. We precisely quantify this gain-cost tradeoff, document frequency-dependent knowledge transfer absent from the RL-trained teacher, and characterize a calibration gap that SFT does not close—identifying both as directions for future work.

## 1 Introduction

Rare diseases collectively affect an estimated 263–446 million people worldwide (Nguengang Wakap et al., 2020; Bruckner-Tuderman, 2021), yet clinical NLP research focuses predominantly on common conditions. ZebraMap (Islam et al., 2026b)—a multimodal resource constructed from 36,131 PubMed case report articles into 69,146 patient-level narratives and 98,038 clinical images across 1,727 rare diseases—provides a principled testbed for diagnostic reasoning on underrepresented conditions.

Deploying large reasoning models in resource-constrained clinical settings requires smaller, distilled models (Hinton et al., 2015; Xu et al., 2024). Standard CoT distillation (Wei et al., 2022) samples teacher traces on unseen inputs for behavioral cloning (Ho et al., 2023; Magister et al., 2023), limiting knowledge transfer to what the teacher already knows without label guidance. We study **hindsight-guided distillation**, where the teacher generates rationales with the correct diagnosis label visible, producing label-anchored reasoning chains that explicitly connect patient evidence to confirmed diagnoses. This is related to STaR (Zelikman et al., 2022) but uses a dedicated external teacher rather than self-generated rationales.

**Trace contamination, instantiated as GT phrases.** Trace contamination from label-visible generation is a known phenomenon (Green et al., 2025; Turpin et al., 2023): when teacher traces contain privileged information, this information can leak into student outputs through pattern-copying. In our setting, this manifests as explicit “ground truth is X” phrases—the most common construction appearing in teacher reasoning traces. Label-visible prompting causes DeepSeek-R1-Distill-LLaMA-8B (Guo et al., 2025) to write this phrase in its transparent reasoning chain, and SFT faithfully copies the structural pattern (Turpin et al., 2023). At inference, Student reproduces the phrase despite no label access, wrong in 89.1% of contaminated cases, causing a 18 percentage point Accuracy collapse. We refer to this specific instantiation as **Ground Truth (GT) hallucination**.

**Regex-based filtering and its limits.** The natural data-centric response is to remove GT reference slots from training traces before SFT. We train two variants on the same 18,388 cases:

- (1) **StudentF**: trained on regex-filtered traces with all “ground truth is X” slots removed. Produces

near-zero GT phrases at inference (3/2,000 traces retain contamination).

- (2) **Student:** trained on raw traces. Serves as an ablation quantifying what unfiltered hindsight distillation produces.

StudentF exceeds the teacher on Accuracy across all metric variants (+0.040,  $p < 0.001$ ). The unfiltered Student, by contrast, does *not* significantly outperform the teacher ( $p = 0.129$ ), demonstrating that filtering is necessary to realise the advantage. The gain of filtering over the unfiltered Student (+0.024,  $p < 0.05$ ) is partly offset because static filtering cannot distinguish labels that would be reproduced correctly at inference from those that would not—and correct labels provide genuine diagnostic value (Student Accuracy on the GT-correct stratum: 0.7703).

## Contributions.

- (1) A study of hindsight-guided distillation on ZebraMap that rigorously quantifies the trace contamination artifact produced by label-visible trace generation, with a filtered variant (StudentF) and an unfiltered ablation (Student).
- (2) An evaluation framework with three Accuracy variants (primary tag-based, teacher upper bound, format-compliant subset) and a Mentioned search-space proxy, all novel for this setting.
- (3) Precise quantification of trace contamination in hindsight-guided distillation: prevalence (33.9%), per-stratum accuracy impact, and the gain–cost tradeoff of static filtering, which eliminates the vast majority of contamination (near-zero at inference) and achieves a significant net gain while incurring a correct-label cost in a subset of cases.
- (4) Evidence consistent with established findings (Yang et al., 2025) that SFT produces frequency-dependent accuracy gains and format imitation without epistemic calibration, motivating investigation of more refined training approaches as future work.

## 2 Related Work

**Chain-of-thought distillation.** Ho et al. (2023) showed zero-shot CoT traces can fine-tune smaller models for multi-step reasoning; Magister et al. (2023) transferred CoT capabilities to sub-100B students; Hsieh et al. (2023) showed joint label-and-rationale supervision outperforms answer-only

distillation; Wadhwa et al. (2024) further analysed what makes CoT augmentation effective in distillation. Chen et al. (2025b) systematically examined how granularity, format, and teacher choice affect CoT distillation, finding that stronger teachers do not always produce better students. Wadhwa et al. (2024) showed that CoT distillation gains can survive shuffled rationales, suggesting that format signal alone can drive improvement. Most pertinently, Yang et al. (2025) showed that SFT-based reasoning improvements arise primarily from format alignment rather than genuine reasoning enhancement. All these work without ground-truth label access at trace generation time. We study the complementary regime where the teacher sees the label, characterizing the resulting artifact.

**Label-conditioned and hindsight rationale generation.** Zelikman et al. (2022) introduced STaR, bootstrapping rationale generation by conditioning on correct answers. Li and Zhang (2026) extended STaR with knowledge grounding. We differ in using a dedicated external teacher whose input contains the ground-truth label explicitly—producing stronger label-anchored reasoning but also contamination artifacts (Green et al., 2025). Our contribution is to precisely quantify the extent and impact of this contamination in a clinical setting.

**Data-centric trace selection.** Recent work shows trace quality filtering improves CoT distillation without RL (Zhuang et al., 2026; Zhou et al., 2023); the DC-CoT benchmark (Zhang et al., 2026) provides a broader framework for understanding data selection strategies. We apply this principle specifically to contamination filtering and show that targeted regex filtering eliminates the vast majority of contamination at inference but yields a bounded net gain due to the correct-label tradeoff.

**LLMs for rare disease diagnosis.** Singhal et al. (2023) demonstrated that large language models can encode substantial clinical knowledge; Segura-Bedmar et al. (2022) showed deep learning methods can identify rare diseases and their manifestations from text. Ao et al. (2025) benchmarked proprietary models on rare disease cases; Young et al. (2025) evaluated GPT-4 on paediatric presentations; Reese et al. (2026) found general-purpose LLMs lag behind specialist tools on phenopacket-driven diagnostics; Wu et al. (2025b) showed knowledge-graph-guided prompting elicits more factual medical reasoning steps. RareArena (Chen

et al., 2026) provides a complementary benchmark; MedR1 (Wang, 2025) distills DeepSeek-R1 (the same teacher model we use) into the medical domain; Wu et al. (2025a) showed that CoT prompting can hurt clinical-task accuracy. Our work complements these by providing a rigorous quantification of the contamination artifact produced by hindsight-guided distillation, at substantially lower inference cost than the teacher.

**SFT vs. reward-based training.** DeepSeek-R1 (Guo et al., 2025) showed GRPO-based RL develops genuine reasoning. Ouyang et al. (2022) established supervised fine-tuning from human feedback as a foundation for instruction-following; Rafailov et al. (2024) proposed Direct Preference Optimisation as a simpler reward-based alternative. Chen et al. (2025a) compared SFT and RL for training R1-like reasoning models; Turpin et al. (2023) identified unfaithful CoT explanations as a signature of format imitation, and Yang et al. (2025) formalised this as format alignment dominating SFT gains. Fu et al. (2026) proposed single-stage SFT+RL integration, finding that SFT alone produces coarse-grained global changes without the fine-grained selective optimisations of RL. Our results are consistent with these characterisations in a clinical diagnostic setting.

### 3 ZebraMap

ZebraMap (Islam et al., 2026b) is a multimodal rare-disease knowledge map built from 1,727 Orphanet diseases and 36,131 PubMed case report articles (yielding 69,146 patient-level cases with standardised fields), along with 98,038 linked clinical images. Case structuring follows a hybrid pipeline: rule-based section detection and regex-based demographic extraction first isolate individual patient narratives, which are then passed to an LLM (Gemini) for standardised field population; a separate RAG component fills missing disease-level fields from retrieved abstracts.

**Splits.** We define our own train/evaluation split from ZebraMap; no canonical split is provided by the dataset. Training: 1,090 diseases, 18,388 cases. Evaluation: 250 diseases  $\times$  8 cases = 2,000 total. All hindsight-guided teacher traces are generated exclusively from training-split cases; no evaluation case is seen during trace generation or fine-tuning. All evaluation diseases also appear in the training split (frequency 1–226, median 17); the

analysis therefore characterizes knowledge transfer at varying training exposure levels, not zero-shot generalization.

## 4 Methodology

### 4.1 Models

**Base.** Qwen-2.5-1.5B-Instruct (Qwen, 2025) (4-bit NF4). Evaluated in two modes: free-form prose (Mentioned baseline) and zero-shot <diagnosis>-tag prompting with no chain-of-thought (Accuracy reference baseline; labelled *Base (tagged)* in tables).

**Teacher.** DeepSeek-R1-Distill-LLaMA-8B (Guo et al., 2025) (GGUF Q4\_K\_M). A CoT distillate of DeepSeek-R1-671B (GRPO-trained); evaluated zero-shot on ZebraMap.

**StudentF (filtered; primary model).** Qwen-2.5-1.5B fine-tuned with QLoRA (Dettmers et al., 2023) (rank 64,  $\alpha=128$ , BnB 4-bit) on 18,388 teacher-generated traces from which a regex filter has removed all explicit “ground truth is X” slots. StudentF produces near-zero GT phrases at inference (3/2,000 = 0.002 contamination rate).

**Student (unfiltered; ablation).** Identical architecture and training to StudentF, but trained on raw teacher traces. Presented alongside StudentF to reveal the structure and magnitude of GT hallucination.

### 4.2 Hindsight Trace Generation

Each trace is generated by prompting the teacher with a case narrative *and* the ground-truth disease label, instructing it to reserve explicit label use for the end of its reasoning. The teacher outputs <think>...</think> followed by <diagnosis>Name</diagnosis>. At inference, both Student and StudentF receive only the case narrative.

This produces label-anchored evidence chains at training time. Because DeepSeek-R1-Distill-LLaMA-8B writes “ground truth is X” inside its transparent deliberative reasoning (Guo et al., 2025), SFT copies this pattern into both trained models. This is a known trace contamination mechanism (Green et al., 2025; Turpin et al., 2023): privileged information present at generation time leaks into outputs through pattern-copying. StudentF’s filter removes these slots from all 18,388 training traces, so the pattern is near-eliminated during its fine-tuning.

### 4.3 Evaluation Metrics

We define four metrics, all novel for this evaluation setting. M1 and M3 are shorthand labels used in tables and figures throughout.

**Mentioned (M1)** checks whether the ground-truth disease name appears anywhere in the full response. It is a search-space proxy only and is never used as an accuracy measure; it credits passing mentions in rejected hypotheses as much as committed diagnoses.

**Accuracy (M3)** is the primary accuracy metric. It extracts the content of the <diagnosis> tag and scores it by bidirectional substring match against a disease synonym list; responses without a tag score 0. Only the model’s committed answer is credited. The Base model is also evaluated in a zero-shot tag-prompting mode with no chain-of-thought, referred to as *Base (tagged)* in Table 1.

**Accuracy, upper bound (M3\_posttag)** is a robustness bound for the teacher. For untagged responses it searches post-think prose; for tagged responses it is identical to M3. This corrects for zero-shot format non-compliance and represents the best-case Accuracy for teacher evaluation.

**Accuracy, format-compliant (M3\_restricted)** is computed only over the  $n=1887$  responses containing a well-formed <diagnosis> tag, removing any confound from format non-compliance.

**Statistical tests.** Primary: clustered permutation (250 disease clusters,  $B=50,000$ ). Confidence intervals: 95% clustered bootstrap ( $B=50,000$ ). Holm–Bonferroni correction within each comparison family.

## 5 Results

### 5.1 Main Accuracy

Table 1 reports all metrics with 95% clustered bootstrap CIs. The primary result is that StudentF achieves Accuracy=0.196, significantly exceeding the teacher (0.155,  $p<0.001$ ) and the unfiltered Student (0.172,  $p<0.05$ ). Critically, the unfiltered Student alone does *not* significantly outperform the teacher ( $p=0.129$ ; Table 2), demonstrating that filtering—not hindsight distillation per se—drives the accuracy gain. The contamination pattern underlying these comparisons is a known consequence of label-visible trace generation (Green et al., 2025); our contribution is to quantify its prevalence, per-stratum accuracy impact, and the net effect of filtering with statistical

rigor.

The teacher’s Mentioned (0.253) also substantially exceeds its Accuracy (0.155) by 0.098 because it frequently names the correct disease in deliberation without placing it in a <diagnosis> tag (0.962 tag compliance). Accuracy (upper bound) (0.163) partially recovers these; even so, StudentF’s Accuracy (upper bound) (0.197) exceeds the teacher’s upper bound.

**Consistency of the filtered advantage.** Three alternative metrics confirm the advantage does not depend on the primary Accuracy metric’s tag requirement. On Accuracy (format-compliant) ( $n=1887$  format-compliant cases): StudentF 0.198 vs. teacher 0.163 ( $p<0.05$ ). On Accuracy (upper bound): StudentF 0.197 vs. teacher 0.163 ( $p<0.05$ ). All pairwise comparisons are summarised in Table 2.

### 5.2 Quantifying Trace Contamination

Trace contamination from label-visible generation is a known mechanism (Green et al., 2025; Turpin et al., 2023). Here it takes the form of “ground truth is X” phrases: the teacher writes them during trace generation because the label is part of its input, and SFT copies the pattern. Student reproduces it at inference despite no label access; StudentF does not, because the pattern was near-eliminated from its training data.

**Contamination cascade.** A GT phrase appears in 33.9% [0.3185, 0.3595] of Student inferences. Of those, 89.1% [0.8646, 0.9156] contain a wrong disease. When the wrong label is present, Student copies it into <diagnosis> 74.5% [0.7079, 0.7810] of the time. Table 3 shows the resulting Accuracy stratification: Student drops from 0.2050 (GT absent) to 0.0265 (GT wrong), a 18 percentage point collapse. Figures 1 and 2 illustrate the cascade proportions and the resulting per-stratum Accuracy.

**Why the filtering gain is bounded.** StudentF suppresses GT contamination to near-zero at inference ( $3/2,000 = 0.002$ ), gaining 0.08 Accuracy over Student on the GT-wrong stratum—the intended benefit. But the GT-correct stratum ( $n=74$ ) reveals the cost: Student Accuracy on that stratum (0.7703) is far above the uncontaminated baseline (0.2050), indicating correct GT hints provide genuine diagnostic value. StudentF forfeits this value ( $-0.27$  Accuracy per case on that stratum). The net ef-

| Metric                                    | Base                    | Student                 | StudentF                       | Teacher                 |
|-------------------------------------------|-------------------------|-------------------------|--------------------------------|-------------------------|
| Mentioned (proxy only)                    | 0.191<br>[0.162, 0.221] | 0.255<br>[0.223, 0.288] | 0.255<br>[0.222, 0.287]        | 0.253<br>[0.220, 0.287] |
| <b>Accuracy</b> (primary)                 | 0.069<br>[0.054, 0.083] | 0.172<br>[0.147, 0.198] | <b>0.196</b><br>[0.170, 0.223] | 0.155<br>[0.133, 0.179] |
| Accuracy, upper bound                     | —                       | 0.174<br>[0.149, 0.200] | <b>0.197</b><br>[0.170, 0.224] | 0.163<br>[0.140, 0.187] |
| Accuracy, format-compliant ( $n = 1887$ ) | —                       | 0.175<br>[0.149, 0.202] | <b>0.198</b><br>[0.171, 0.226] | 0.163<br>[0.139, 0.187] |
| <diagnosis> present                       | —                       | 0.987                   | 0.991                          | 0.962                   |
| Mean response length (chars)              | 1950                    | 4098                    | 3753                           | 3421                    |

Table 1: All metrics for 2,000 cases across 250 diseases. 95% clustered bootstrap CIs shown below each point estimate. **Accuracy is the primary metric.** Mentioned is a search-space proxy and is never used as an accuracy measure. Base (tagged): base model in zero-shot tag-prompting mode, no chain-of-thought.

| Comparison                   | Metric    | RD [95% CI]             | Cohen’s $h$ [95% CI]    |
|------------------------------|-----------|-------------------------|-------------------------|
| <i>Primary (M3/Accuracy)</i> |           |                         |                         |
| StudentF vs. Teacher         | Acc.      | +0.040 [+0.019, +0.061] | +0.105 [+0.052, +0.160] |
| Student vs. Teacher          | Acc.      | +0.016 [−0.004, +0.037] | +0.045 [−0.012, +0.101] |
| StudentF vs. Student         | Acc.      | +0.024 [+0.006, +0.043] | +0.061 [+0.010, +0.115] |
| StudentF vs. Base            | Acc.      | +0.127 [+0.105, +0.150] | +0.386 [+0.327, +0.446] |
| <i>Conservative bounds</i>   |           |                         |                         |
| StudentF vs. Teacher         | Acc. (UB) | +0.033 [+0.012, +0.056] | +0.088 [+0.033, +0.148] |
| StudentF vs. Teacher         | Acc. (FC) | +0.035 [+0.013, +0.058] | +0.093 [+0.035, +0.152] |

Table 2: Pairwise comparisons. RD = risk difference (A−B); Cohen’s  $h$  = effect size on arcsine scale. All CIs: 95% clustered bootstrap;  $p$ -values: Holm–Bonferroni adjusted clustered permutation. StudentF exceeds the teacher under every Accuracy variant ( $p < 0.001$  primary;  $p < 0.01$  conservative bounds). Student alone does not significantly outperform the teacher ( $p = 0.129$ ). StudentF vs. Student ( $p = 0.015$ ) is significant, confirming that filtering—not distillation alone—drives the gain; see Section 5.2.

fect is a +0.024 Accuracy gain for StudentF overall ( $p < 0.05$ ), indicating that filtering is beneficial on balance.

Crucially, the advantage is not limited to contamination removal: on the 0.2050 GT-absent stratum ( $n = 1322$ ), StudentF (0.2194) also significantly outperforms the Teacher (0.1793,  $p_{\text{adj}} = 0.009$ ), while Student does not ( $p_{\text{adj}} = 0.17$ ). Filtering therefore improves not only the contaminated cases but also produces better clean-case reasoning than unfiltered SFT.

This gain is structurally bounded. Training-time labels are always correct, so a static filter cannot know which phrases will be reproduced correctly at inference.

## 6 Analysis

### 6.1 Training Frequency and Knowledge Transfer

StudentF Accuracy scales significantly with training frequency (Spearman  $\rho = +0.242$ ,  $p < 0.01$ ; OLS slope +0.058,  $p < 0.05$ ). The teacher shows no significant frequency dependence ( $\rho = +0.029$ ,  $p = 0.65$  (ns)). This dissociation is consistent with the student’s gains reflecting knowledge transfer from training traces rather than model capacity, though it is also consistent with format alignment and memorisation of common disease presentations (Yang et al., 2025; Wadhwa et al., 2024). The StudentF–Teacher gap grows from −0.016 in Q1 to +0.121 in Q4 ( $\rho$  for SF–T gap: +0.300,  $p < 0.001$ ). At Q1 (median 7 training cases) Teacher marginally leads StudentF (−0.016) and all three models are statistically indistinguishable—SFT cannot create knowledge absent from training.

| Stratum (defined by Student’s reasoning chain) | <i>n</i> | Student Accuracy           | StudentF Accuracy          | Teacher Accuracy           |
|------------------------------------------------|----------|----------------------------|----------------------------|----------------------------|
| No GT phrase                                   | 1322     | 0.2050<br>[0.1735, 0.2373] | 0.2194<br>[0.1891, 0.2504] | 0.1793<br>[0.1502, 0.2094] |
| GT phrase, label correct                       | 74       | 0.7703<br>[0.6712, 0.8611] | 0.5000<br>[0.3803, 0.6164] | 0.2973<br>[0.1867, 0.4143] |
| GT phrase, label wrong                         | 604      | 0.0265<br>[0.0142, 0.0408] | 0.1060<br>[0.0810, 0.1329] | 0.0960<br>[0.0721, 0.1219] |

Table 3: Accuracy stratified by GT phrase presence in the Student’s reasoning chain. Strata are defined by Student behaviour; StudentF and Teacher values cover the same case subsets for comparability. StudentF produces near-zero GT phrases at inference (3/2,000); Teacher produces none. Clustered permutation test (Student: GT-wrong vs. GT-absent):  $p < 0.001$ .

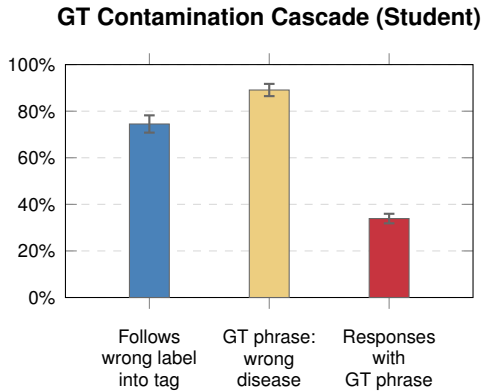


Figure 1: GT contamination cascade for the unfiltered Student. Of 2,000 inferences, 33.9% [0.3185, 0.3595] contain a GT phrase; of those, 89.1% [0.8646, 0.9156] name the wrong disease; of those, 74.5% [0.7079, 0.7810] are copied into the <diagnosis> tag. Error bars: 95% clustered bootstrap CIs.

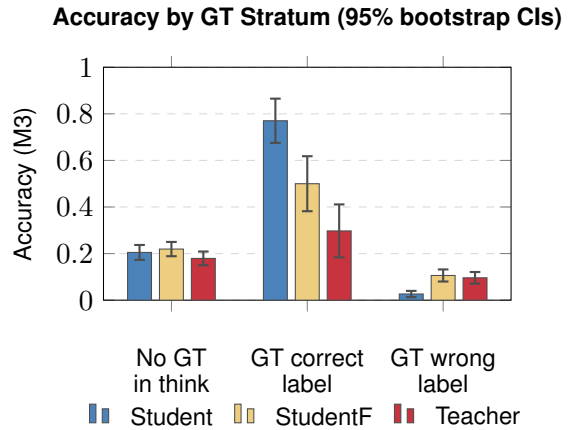


Figure 2: M3 accuracy stratified by GT phrase presence in Student’s reasoning chain. StudentF: zero GT phrases at inference (filter removed all from training). Teacher: zero GT phrases (zero-shot; no label access). Clustered permutation, Student GT-wrong vs. GT-absent:  $p < 0.001$ .

## 6.2 Reasoning Style and Epistemic Calibration

Table 5 shows reasoning style by model. All three models think longer, self-correct more, and express more uncertainty when wrong—a pattern captured by the SC ratio—but the absolute rates differ dramatically across models.

Filtering has an unexpectedly strong suppressive effect on StudentF’s reasoning behaviour. StudentF self-corrects at 0.15 per response, significantly below the teacher’s 0.28 ( $p < 0.001$ ) and far below Student’s 0.83. Explicit uncertainty follows the same pattern: StudentF 2.12 vs. Teacher 4.09 vs. Student 4.53. Both effects are consistent with GT-label contamination causing extra counter-reasoning in Student traces (“wait—the ground truth says X, but...”); removing that contamination from training data suppresses the associated hedging behaviour in StudentF.

The most salient Teacher-vs-Student contrast is in confident assertions. Teacher commits

assertively at 0.67 per response vs. 0.31 for Student—a  $2.1 \times$  difference ( $p < 0.001$ ), consistent with RL training toward committed, evaluable answers (Chen et al., 2025a; Turpin et al., 2023). StudentF’s filtering partially closes this gap (0.51,  $p < 0.001$  vs. Teacher), suggesting that removing GT contamination also removes the conditionally uncertain reasoning that suppresses assertion confidence. All models show a modest positive gap in confident assertions on correct vs. wrong answers with similar relative magnitude (Li et al., 2024).

The SC ratio is  $> 1$  for all three models directionally, but StudentF’s CI [0.9646, 1.9537] straddles 1.0, meaning the elevation of self-correction on wrong responses is not statistically distinguishable from noise for StudentF specifically. Student and Teacher show more reliable miscalibration ([1.2969, 1.7884] and [1.0984, 2.0598] respectively). This pattern of format imitation without epistemic calibration is consistent with findings

| Q      | Freq | Stud Acc.                      | SF Acc.                        | Teach Acc.              |
|--------|------|--------------------------------|--------------------------------|-------------------------|
| Q1     | 7    | 0.104<br>[0.068, 0.143]        | 0.133<br>[0.096, 0.174]        | 0.148<br>[0.107, 0.191] |
| Q2     | 13   | 0.115<br>[0.075, 0.159]        | 0.139<br>[0.095, 0.188]        | 0.137<br>[0.091, 0.188] |
| Q3     | 25   | 0.185<br>[0.141, 0.234]        | 0.214<br>[0.159, 0.270]        | 0.159<br>[0.113, 0.208] |
| Q4     | 59   | <b>0.286</b><br>[0.226, 0.349] | <b>0.298</b><br>[0.238, 0.359] | 0.177<br>[0.131, 0.226] |
| $\rho$ |      | +0.306**                       | +0.242*                        | +0.029 (ns)             |
| OLS    |      | +0.064**                       | +0.058*                        | -0.000 (ns)             |

Table 4: Mean per-disease Accuracy by training-frequency quartile with 95% clustered bootstrap CIs. The StudentF–Teacher gap grows from  $-0.016$  in Q1 (where Teacher marginally leads) to  $+0.121$  in Q4. Teacher shows no frequency dependence. \*\* $p < 0.01$ , \* $p < 0.05$ , ns:  $p \geq 0.05$ .

that SFT produces format alignment rather than genuine adaptive reasoning (Yang et al., 2025; Jiang et al., 2026).

## 7 Discussion

**What hindsight-guided distillation with contamination filtering achieves.** StudentF demonstrates that a 1.5B model trained via SFT on 18,388 hindsight-guided traces can achieve a narrow accuracy advantage over its 8B teacher on ZebraMap’s rare-disease diagnosis task ( $+0.040$  M3,  $p < 0.001$ ). Notably, the unfiltered Student alone does *not* significantly outperform the teacher ( $p = 0.129$ ): it is the combination of hindsight distillation *and* contamination filtering that drives the gain. The advantage is concentrated in better-represented diseases; at Q1 (median 7 training cases) Teacher marginally leads StudentF ( $-0.016$ ), with all three models statistically indistinguishable, consistent with the student’s gains reflecting knowledge transfer from well-represented training cases (or format alignment with those patterns) rather than model capacity. Deploying a 1.5B student at inference costs far less than serving the 8B teacher.

**The limits of static filtering.** Filtering correct GT labels alongside incorrect ones is unavoidable with a static regex: the filter cannot observe inference-time behaviour. Our results quantify the cost: the GT-correct stratum ( $n = 74$ ) achieves Student Accuracy = 0.7703, far above the uncontaminated baseline (0.2050). Removing these hints reduces StudentF Accuracy on that stratum to 0.5000—a partial offset against the gains on the GT-wrong stratum. Despite this cost, filtering yields a

statistically significant net gain overall ( $p < 0.05$ ). More sophisticated selection approaches—such as training a classifier to predict inference-time reproduction accuracy, or optimistic consistency filtering (Wang et al., 2023a,b)—may reduce the correct-label cost further, but the fundamental limitation persists: the filter never observes the inference error it is trying to prevent.

**SFT without reinforcement learning.** StudentF acquires the teacher’s reasoning format (0.991 tag adherence) and frequency-sensitive diagnostic knowledge, but not epistemic calibration: its confident assertions do not track correctness, and its SC ratio (1.3365) remains above one without genuine uncertainty meaning. This pattern of format imitation without adaptive calibration is consistent with findings that SFT produces format alignment rather than genuine reasoning (Yang et al., 2025). Jiang et al. (2026) further found that medical CoT distillation degrades factual reliability even when accuracy improves, suggesting that accuracy gains alone do not indicate trustworthy reasoning. A reward signal that penalises wrong diagnoses at inference would simultaneously address both failures—directly improving diagnostic accuracy, and specifically discouraging hallucinated GT phrases (since following a wrong GT label into <diagnosis> produces a reliably penalised outcome). GRPO-based training has shown this effect in general reasoning settings (Guo et al., 2025; Shao et al., 2024); verbal reinforcement approaches such as Reflexion (Shinn et al., 2023) offer a complementary direction for iterative self-correction without full RL.

**Disease-vocabulary confound and the dataset ceiling.** BM25 retrieval at  $k = 1$  achieves 0.3935 label-classification accuracy on our evaluation set—more than twice the best model’s M3 Accuracy (0.196)—despite using no medical knowledge beyond token counts. Low ROUGE-L overlap between evaluation cases and their BM25 matches (mean 0.1483, median 0.1385) confirms that BM25 does not succeed via text copying; the signal is disease-vocabulary leakage: cases concerning the same rare disease share distinctive symptom clusters, syndrome names, and genetic markers that act as near-unique identifiers. A concurrent study on ZebraMap by Alkhathlan et al. (Islam et al., 2026a) independently corroborates this mechanism: using a hybrid retrieval-and-classification pipeline with no LLM reasoning, they find that a class-balanced TF–IDF classifier is their strongest com-

| Metric                           | Student                    | StudentF                   | Teacher                    |
|----------------------------------|----------------------------|----------------------------|----------------------------|
| Think length (chars)             | 3253                       | 2984                       | 3202                       |
| Self-corrections                 | 0.83                       | 0.15                       | 0.28                       |
| Expl. uncertainty                | 4.53                       | 2.12                       | 4.09                       |
| Confident assertions             | 0.31                       | 0.51                       | 0.67                       |
| SC ratio ( $\times/\checkmark$ ) | 1.5143<br>[1.2969, 1.7884] | 1.3365<br>[0.9646, 1.9537] | 1.4650<br>[1.0984, 2.0598] |

Table 5: Mean reasoning-style features (overall) and SC ratio (95% clustered bootstrap CI). All models hedge more when wrong; SC ratio  $> 1$  directionally for all three, though StudentF’s CI [0.9646, 1.9537] straddles 1.0.

ponent (MRR 0.4200, Recall@10 0.6279), with BM25 serving as a complementary but individually weaker signal. Their analysis also applies grouped splitting by source article—a control not present in our disease-level split—suggesting that source-article overlap among training diseases may further inflate the BM25 upper bound. Together, these results reinforce that absolute accuracy figures on ZebraMap at this model scale reflect the disease-vocabulary confound as much as diagnostic reasoning; the relative ordering of models, which is the primary object of our analysis, remains the more interpretable quantity.

## 8 Conclusion

We studied hindsight-guided rationale distillation for rare-disease diagnosis, training a 1.5B student on chain-of-thought traces generated by an 8B teacher with ground-truth label access. Trace contamination from label-visible generation is a known mechanism (Green et al., 2025); we precisely quantify its prevalence (33.9% of Student inferences), accuracy impact (18pp collapse when the leaked label is wrong), and the trade-offs of static filtering in this setting. The filtered variant StudentF achieves near-zero contamination (3/2,000) and a narrow accuracy advantage over the teacher ( $p < 0.001$ ), while the unfiltered Student does not significantly outperform the teacher ( $p = 0.129$ ). Filtering yields a statistically significant net gain ( $p < 0.05$ ), though a correct-label cost partially bounds the benefit. These measurements provide a rigorous characterization of this contamination artifact in a clinical diagnostic context, connecting to broader concerns about trace privacy and SFT faithfulness.

## Limitations

**Frequency-distribution sensitivity.** StudentF’s aggregate advantage over the Teacher (+ + 0.040 M3) is entirely concentrated in high-exposure dis-

eases: in Q1 (median 7 training cases) Teacher leads StudentF by 1.6 pp, with all models statistically indistinguishable. The aggregate result therefore depends on the frequency distribution of the evaluation set; an evaluation set weighted toward low-frequency diseases would reduce or reverse the advantage.

**Performance ceiling.** Even the Teacher achieves only 0.155 M3 Accuracy. The structural ceiling analysis finds that 46% of eval diseases (115/250) have max M3\_posttag below 0.25 across all models. The mean Spearman correlation between per-disease M3 accuracy rankings, averaged across all  $\binom{4}{2} = 6$  model pairs, is  $\bar{\rho} = 0.623$ —indicating moderate difficulty concordance across models. This shared difficulty profile suggests the binding constraint is the task itself at this model scale, not model-specific capacity; distillation gains accrue on top of a ceiling that affects teacher and students equally.

**Trace contamination detection is a lower bound.** The contamination analysis relies on a single family of regex patterns targeting “ground truth is X” constructions. Alternative phrasings not captured by this pattern go undetected, so the measured 33.9% Student contamination rate is a lower bound on true GT exposure in the reasoning chain.

**Static filter cost.** The regex filter used to produce StudentF removes GT slots without observing which labels would be reproduced correctly at inference. On the GT-correct stratum ( $n = 74$ ), StudentF Accuracy drops by 27 pp relative to Student, as correct hints that would have been followed faithfully are excised along with incorrect ones. This cost is irreducible with a static filter; eliminating it requires an inference-time signal.

**Pre-distillation baseline.** We evaluate the Base model only in a zero-shot tag-prompting mode

without chain-of-thought, not in the same CoT format used by the trained students. As Kajitsuka et al. (2026) demonstrate, comparisons between distilled students and teachers without proper pre-distillation baselines can be misleading; distillation can sometimes degrade baseline performance.

**Mechanism ambiguity.** StudentF’s accuracy gains over the teacher are consistent with both genuine knowledge transfer from hindsight-guided traces and format alignment (Yang et al., 2025). Wadhwa et al. (2024) showed that CoT distillation gains can survive shuffled rationales, suggesting that format signal alone can drive improvement. Our measurements quantify the net accuracy effect but cannot distinguish the underlying mechanism.

**Quantisation.** Student and StudentF are evaluated at 4-bit NF4 while the Base (tagged) reference runs at full floating-point precision. This asymmetry is conservative for fine-tuned models: any advantage over Base is observed despite a quantisation disadvantage. Cross-format effects between student NF4 and teacher Q4\_K\_M are unmeasured.

**Statistical clustering.** Tests and CIs cluster by disease (250 clusters, 8 cases each). ICC(1) values of 0.17–0.20 empirically confirm that disease-level clustering is warranted; however, strict exchangeability does not hold—diseases that are hard for one model tend to be hard for all (same  $\bar{\rho}$ =0.623 concordance across model pairs), violating the assumption that clusters are interchangeable draws from a uniform difficulty distribution. CIs are therefore conservative approximations rather than exact intervals.

**Training randomness.** A single Student and a single StudentF are reported. Evaluation CIs are clustered over disease strata; variance across training runs is not quantified.

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|     | <b>A Error Taxonomy</b>                                                                   |     |
|     | <b>Category definitions (priority cascade).</b> Each                                      |     |
|     | response is assigned to exactly one category, evalu-                                      |     |
|     | ated in the order listed below. The first matching                                        |     |
|     | criterion determines the label.                                                           |     |
|     | (1) <b>Correct (M3):</b> The <diagnosis> tag con-                                         |     |
|     | tent matches the ground-truth disease via bidi-                                           |     |
|     | rectional synonym match (the primary Accu-                                                |     |
|     | racy metric).                                                                             |     |
|     | (2) <b>Format failure — recovered:</b> No valid tag,                                      |     |
|     | but the correct disease appears in post-think                                             |     |
|     | prose (captured by Accuracy, upper bound).                                                |     |
|     | (3) <b>GT-poisoned:</b> Response contains a “ground                                       |     |
|     | truth is X” phrase with the wrong disease,                                                |     |
|     | which is then copied into the <diagnosis>                                                 |     |
|     | tag. Student-exclusive artifact.                                                          |     |
|     | (4) <b>Revised away:</b> The correct disease appears                                      |     |
|     | in the think chain (M1-correct) but the model                                             |     |
|     | self-revised away from it before committing to                                            |     |
|     | a wrong <diagnosis> tag.                                                                  |     |
|     | (5) <b>Near-miss:</b> The predicted disease is wrong by                                   |     |
|     | M3, but its embedding has cosine similarity                                               |     |
|     | > 0.7 with the true disease embedding (seman-                                             |     |
|     | tically very close).                                                                      |     |
|     | (6) <b>Weak related:</b> Cosine similarity $0.5 < \cos \leq$                              |     |
|     | 0.7 to the true disease (semantically related but                                         |     |
|     | clearly distinct).                                                                        |     |
|     | (7) <b>Wrong — same ICD-10 category:</b> Prediction                                       |     |
|     | and ground truth are in the same ICD-10 chap-                                             |     |
|     | ter but cosine similarity $\leq 0.5$ .                                                    |     |
|     | (8) <b>Wrong — different ICD-10 category:</b> Predic-                                     |     |
|     | tion and ground truth are in different ICD-10                                             |     |
|     | chapters; both diseases have ICD-10 codes.                                                |     |
|     | (9) <b>Both uncategorized:</b> Neither the predicted                                      |     |
|     | nor the true disease has an assigned ICD-10                                               |     |

code. Occurs for very rare diseases not yet in the ICD-10 registry.

- (10) **Prediction uncategorized:** The predicted disease has no ICD-10 code, but the true disease does.
- (11) **True uncategorized:** The true disease has no ICD-10 code, but the predicted disease does.
- (12) **Format failure — unrecoverable:** No valid tag and no recoverable answer in post-think prose.

Categories (9)–(11) collectively cover all wrong predictions where at least one of the two diseases (predicted or true) lacks an ICD-10 code, reflecting the prevalence of extremely rare diseases in ZebraMap that are not yet assigned ICD-10 codes. These constitute a substantial fraction of wrong-answer cases for all models.

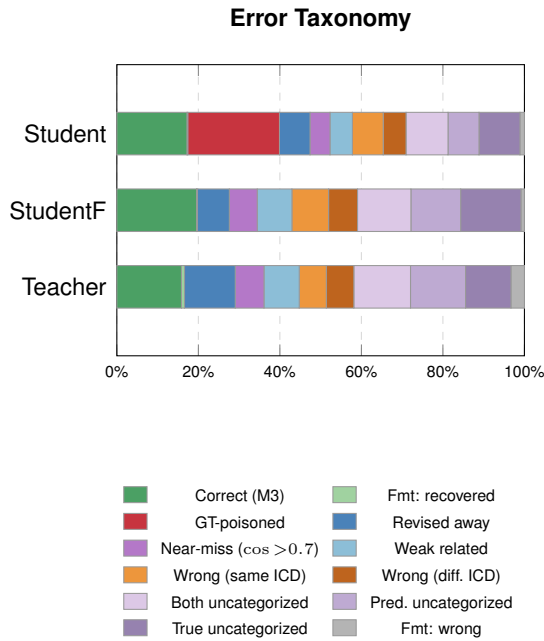


Figure 3: Error taxonomy following the pipeline A.6 priority cascade (see Appendix A for category definitions). GT-poisoned is exclusive to Student (22.5%). ICD-10 uncategorized buckets (lavender shades): *Both uncategorized* = neither predicted nor true disease has an ICD-10 code; *Pred. uncategorized* = predicted disease lacks an ICD-10 code but true disease has one; *True uncategorized* = true disease lacks an ICD-10 code but predicted disease has one. These cover rare diseases not yet assigned ICD-10 codes and constitute a large share of wrong-answer cases.

## B Bidirectional Match Parameter Sweep

The primary Accuracy metric uses a bidirectional synonym-list match with no minimum character constraint ( $mc = \text{None}$ ; default). Figure 4 shows StudentF and Teacher Accuracy across  $mc = 0$ –10. The StudentF advantage is directionally consistent at every setting ( $SF > T$  across all 22 tested values including  $mc = \text{None}$ ). The  $SF - T$  gap grows as substring leniency increases: from 0.0220 at  $mc = 0$ –3 to 0.0255–0.0285 at  $mc = 5$ –7, peaking at 0.0400 at the exact/synonym default. At low  $mc = 0$ –4, any-length substring matches inflate all three models’ scores equally, compressing the absolute gap while leaving the ordering unchanged.

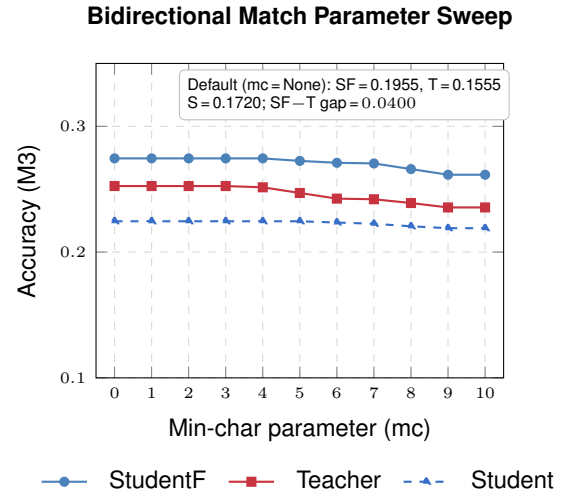


Figure 4: All three models’ Accuracy (M3) across bidirectional match strictness levels ( $mc = 0$ –10). The StudentF advantage over Teacher is directionally consistent at every setting; the  $SF - T$  gap grows from 0.0220 at  $mc = 0$  to 0.0285–0.0255 at  $mc = 5$ –7, peaking at 0.0400 at the exact/synonym default ( $mc = \text{None}$ , dashed reference line in annotation). At low  $mc = 0$ –4, any-length substring matches inflate all three models’ scores equally, compressing the detectable gap. The default setting is used throughout the main paper.

## C Metric and Style Feature Definitions

**Accuracy (M3).** A response is correct if the content of the `<diagnosis>` tag matches the ground-truth disease name via bidirectional substring match against a synonym list: either the tag content contains the ground-truth name, or vice versa (default;  $mc = \text{None}$ ). Responses with no `<diagnosis>` tag score 0.

**Mentioned (M1).** The ground-truth disease name appears anywhere in the full response text

(case-insensitive). This credits passing mentions in rejected hypotheses and is reported for completeness only.

**Self-correction count.** Number of self-correction phrases (e.g., “however”, “on the other hand”, “wait”, “actually”, “let me reconsider”) per response, counted in the <think> chain.

**Explicit uncertainty count.** Number of hedging phrases indicating explicit uncertainty (e.g., “I’m not sure”, “it’s possible that”, “could be”, “uncertain”) per response.

**Confident assertions.** Number of assertive commitment phrases (e.g., “the diagnosis is”, “this is clearly”, “the patient has”, “consistent with”) per response.

**SC ratio.** Ratio of mean self-correction count on incorrect responses to mean self-correction count on correct responses. A ratio of 1.0 indicates equal hedging regardless of correctness; higher values indicate elevated hedging on wrong answers.

## **D Reproducibility Details**

|                                                 |                                                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------|
| <i>Dataset — ZebraMap (Islam et al., 2026b)</i> |                                                                               |
| Eval set                                        | 250 diseases $\times$ 8 cases = 2,000                                         |
| Train set                                       | 1,090 diseases, 18,388 cases (4 quality filters)                              |
| Splits                                          | Raw CoT $\rightarrow$ Student; GT-redacted CoT $\rightarrow$ StudentF         |
| <i>Hindsight trace generation</i>               |                                                                               |
| Teacher                                         | DeepSeek-R1-Distill-LLaMA-8B (Guo et al., 2025) (GGUF Q4_K_M) via llama.cpp   |
| Input                                           | Case narrative + ground-truth label                                           |
| Params                                          | max 4,096 tokens; temperature 0.6                                             |
| <i>Student training</i>                         |                                                                               |
| Base                                            | Qwen-2.5-1.5B-Instruct (Qwen, 2025) (Unsloth, 4-bit NF4)                      |
| LoRA                                            | $r=64$ , $\alpha=128$ , all 7 projection layers                               |
| Schedule                                        | 3 epochs; batch 32 ( $4 \times 8$ grad. accum.); lr $2 \times 10^{-4}$ cosine |
| Other                                           | max 8,192 tokens; Paged AdamW 8-bit; wd $10^{-3}$ ; seed 3,407                |
| <i>Evaluation</i>                               |                                                                               |
| Prompt                                          | Zero-shot; case narrative only                                                |
| Teacher                                         | llama.cpp; max_tokens 4,096; temp 0.6                                         |
| Students                                        | Transformers/Unsloth; max_new_tokens 2,048; temp 0.6; top- $p$ 0.9            |
| Base                                            | Full floating-point (conservative for fine-tuned models)                      |
| BM25                                            | $k_1=1.5$ , $b=0.75$ , $k=1$ ; accuracy = 0.3935                              |
| <i>Statistical methodology</i>                  |                                                                               |
| Clusters                                        | 250 diseases $\times$ 8 cases each                                            |
| Perm. test                                      | $B=50,000$ ; two-tailed; Holm–Bonferroni corrected                            |
| Bootstrap                                       | $B=50,000$ ; 95% percentile; effects: RD, Cohen’s $h$ , OR                    |
| ICC(1)                                          | Shrout & Fleiss (1979) Case-1; disease as grouping factor                     |
| Freq.                                           | OLS on $\log(1+\text{freq})$ and Spearman $\rho$                              |
| Calibration                                     | Grouped $k$ -fold logistic regression; OOF AUC per model                      |
| Stats file                                      | Auto-generated to generated_stats.tex; \inputted at compile time              |

Table 6: Reproducibility settings.